Sahaj Bhadja

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Summary

Software Engineer pursuing an M.S. in Computer Science at USC with 2+ years building **Python/C++ tooling**, robust **ML data pipelines**, and reliable **data curation/validation workflows** for autonomy-focused, safety-critical engineering systems.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University **Gandhinagar, India**
B.Tech. in Computer Science and Engineering *November 2020 – May 2024*
GPA: 3.88/4 Coursework: OOP, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built **Python** workflow services and data interfaces for **500+ users**, reducing latency from **1.2s to 350ms** through query optimization while improving script reliability and backend observability across **Linux**-based systems.
- Implemented fault-tolerant orchestration for **5K+ monthly jobs** with validation safeguards and rollback checks, improving **data availability** and accelerating issue triage across distributed services and cross-functional engineering workflows.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Developed **React** and **Flask** features handling **10K+ daily requests**, translating requirements into reusable APIs and components while strengthening data quality checks in production-facing grading and evaluation pipelines.
- Improved release velocity with **CI/CD** controls and concurrent processing, reducing turnaround time by **60%** while maintaining quality through automated validation checks and structured defect-triage practices with stakeholders.

USC AI Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **Python** experimentation pipelines over **100K+ trajectory logs**, enabling rigorous behavior analysis and reproducible benchmark runs that supported data curation for nominal and rare scenario evaluation tasks.
- Produced technical summaries and debugging diagnostics that converted research findings into implementation plans, improving **defect isolation speed** and strengthening collaboration across ML, perception, and software infrastructure teams.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API** and **ETL** microservice workflows with OOP, AWS Glue, and S3, improving response times by **30%** while supporting scalable integrations and dependable data exchange across enterprise environments.
- Maintained **20+** unit and integration suites using Jenkins, PyTest, and Mockito across **Linux/Windows** environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%** through automation.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **REST API** and **WebSocket** modules supporting eight workflows and **2M+ synchronized records**, improving extensibility and reliability while coordinating integration constraints with engineering and operations teams.
- Automated deployments and testing with **Docker** and **Jenkins**, reducing deployment time by **50%** and improving release consistency through script-driven checks, reproducible environments, and robust rollback mechanisms.

Skills

Languages: Python, C++, Java, JavaScript, TypeScript, SQL, Bash
Data Engineering: ETL Pipelines, Data Curation, Data Validation, Data Balancing, Large-Scale Logs
Backend & APIs: Flask, FastAPI, REST APIs, WebSockets, Microservices, Distributed Workflows
Cloud & Infrastructure: AWS (S3, Glue, ECS), Azure, Docker, Kubernetes, Linux/UNIX
Testing & Reliability: Script Testing, Unit/Integration Testing, Troubleshooting, CI/CD, Agile/Scrum, Faiss

Projects

Big-Brain | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*

- Built a semantic document retrieval system using **FAISS** and transformer embeddings over **50K+ documents**, enabling high-quality data retrieval and structured curation workflows that improved downstream model evaluation consistency.

Linguistix | *Python, AWS, Docker, CI/CD*

- Developed and deployed a containerized **RESTful API** on AWS CodePipeline and ECS/Fargate with automated **CI/CD** and validation checks, achieving **87% accuracy** and production-grade reliability under sustained workloads.

Chat2SQL | *Java, TypeScript, SQL, Docker, Maven*

- Built an AI-assisted SQL workspace supporting **10+ databases**, improving query authoring and schema exploration through structured data access patterns, repeatable validation flows, and reliable multi-database pipeline behavior.